Chotu AI Platform Design

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Short answer

Chotu should not be one clever model. It should be five small layers, and the chat model is only the mouth. Most of what you want (knowing the backstory, having the latest UPSC info, pressing the right button) comes from memory, a daily knowledge pipeline and a separate decision step, not from a smarter LLM.

Job	Recommended pick	Why	Cost
Chotu's voice in chat	DeepSeek V4.1 Flash 1017 (`deepseek-flash`), after a bake-off against today's NVIDIA route	Confirmed live yesterday on DeepSeek's own pricing page; very cheap with caching; fast enough	$0.15 / $0.60 per 1M tokens off-peak, cache hits $0.003
Latest UPSC info	DeepSeek Search and Fetch in a daily job, never inside a live chat	Facts get checked once, then every student gets the same approved notes	$0.75 per 1,000 calls; free until 1 Nov 2026
Triggers that get each student to act	Cloudflare Clef-flash on Workers AI, picking which trigger from the playbook fires for whom, and when	Same idea as vega-U, but hosted on the platform the server already runs on	$0.09 per 1M input tokens, output free, about 39 ms
Spoken Chotu	Sarvam Bulbul V4 Flash for Hindi, Telugu and Indian English	Telugu is confirmed; under 250 ms to first audio	₹2 per 1,000 characters (₹1 until 12 Nov)
Reading PDFs and photos	Indra OCR for English PDFs and printed scans; Hindi and Telugu wait for later	Fast and free on a CPU; handwritten answers will still need a vision model	Free under $20M revenue

The three biggest changes from today:

1. **Chotu always has notes.** The daily pipeline fills the approved library, so the cautious rule you chose stops producing "I don't have notes on that".

2. **A Journey Ledger replaces chat memory.** Chotu reads a short, editable story card built from events, and every line links to the chat where the student said it.
3. **Voice is the expensive part, not the brain.** A spoken reply costs about 30 times a text reply, so voice goes on shared, pre-recorded moments and on tap, not on every message.

Devil's advocate: what you named

Every link was opened on 18 Oct 2026. Two of your picks hold up, two are good ideas in the wrong package, and three are for later. I couldn't read X directly because it asks for a sign-in, so recent releases come from web search instead.

Pick	Verified facts	The case against	Verdict
DeepSeek V4.1 Flash 1017	Real: `deepseek-flash` now serves V4.1-Flash-1017. 1M context. Off-peak $0.15 / $0.60, peak $0.30 / $1.20, cache hit $0.003. The September V4.1 Flash measured 163 to 227 tokens/s with about 1.3 s to first token (Artificial Analysis)	It is one day old, with no independent scores for 1017 and no Telugu evidence. Our 15 Oct test of V4.1 Flash on NVIDIA's free API timed out. DeepSeek's own API runs in China, which matters for students' emotional messages under India's DPDP rules. Not free.	**Use, after a bake-off.** Sarvam also hosts DeepSeek V4.1 Flash in India at ₹25 / ₹105 per 1M tokens (Sarvam pricing), which answers the hosting worry at roughly peak price.
Fish Audio Drama 3	Released 13 Oct. 35B open-weight model, 83 languages, directed with tags like `<whisper>` and `[warmly]`. $10 per 1M characters until 12 Nov, then $50	Hindi and Telugu are not named or tested anywhere. At list price it costs about twice Bulbul. The licence needs a paid deal above $500k yearly revenue. A misspelled model header silently falls back to an older voice.	**Only for pre-recorded English narration**, where directing the delivery is worth it. Not for live Chotu.

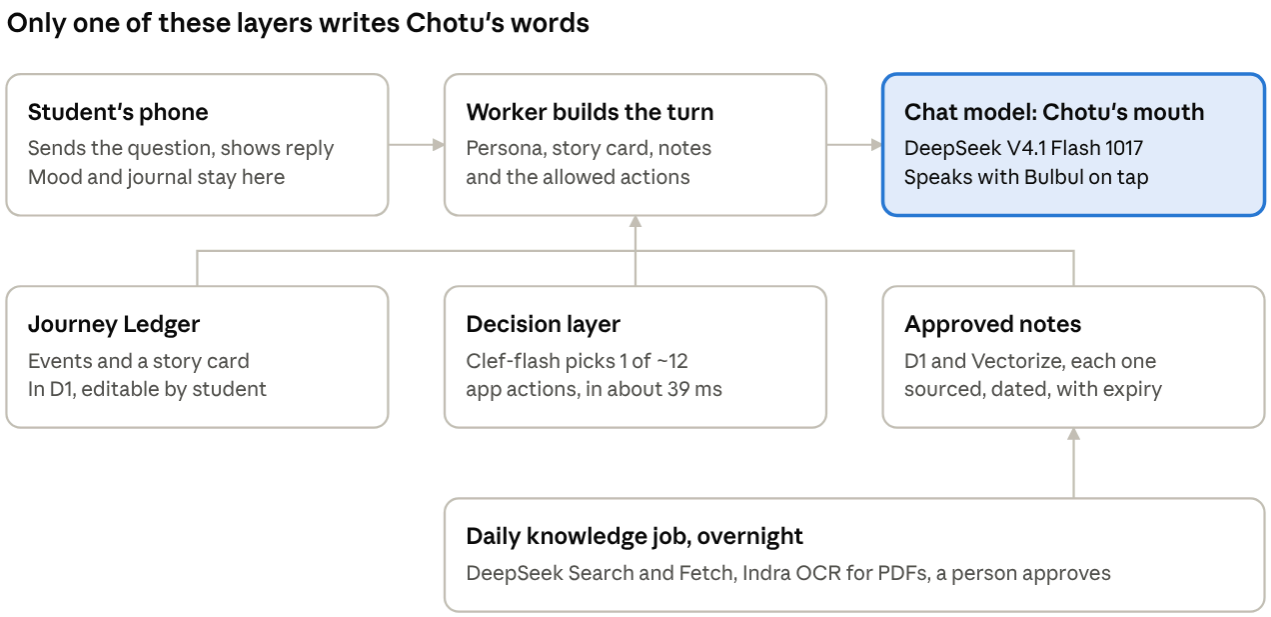
Pick	Verified facts	The case against	Verdict
Sarvam Bulbul V4 Flash	Launched 30 Sep. 11 Indian languages including Telugu, 83 voices, under 250 ms to first audio, streaming over WebSocket. ₹20 per 10,000 characters	Still costs real money per spoken reply, and voice quality in Telugu needs a native listener's check.	**Use** for every live and Indic voice.
DeepSeek Search	Search plus Fetch (page to Markdown), with `site:`, `lang:` and date filters. $0.75 per 1,000 successful calls, 5 per second. Free through Claude Code and similar until 1 Nov	Live search inside a chat pulls in coaching ads and unchecked claims, and adds seconds to every answer.	**Use, but only in the daily knowledge job**, where results get checked before students see them.
vega-U	A 14.3M decision head that plugs into small frozen models such as Qwen3.5-0.8B. Apache 2.0. 0.77 to 0.81 accuracy on its own tests	One author, 2 likes, nobody hosts it, and a wrong nudge to a struggling student is costly.	**Right idea, wrong package.** Use Cloudflare Clef-flash for the same job now. Revisit vega-U to run private decisions on the phone.
Indra OCR	1M-parameter OCR, about 18 MB, CPU only, 230 ms a page. Reads any script's text layer in digital PDFs; Devanagari scans are supported	Telugu scans "come out wrong" and handwriting is unsupported, which rules out photos of handwritten Mains answers. Free only under $20M revenue. It needs Python, so it can't run inside the Worker.	**Use for digital PDFs** (NCERT, PIB, Yojana) in the daily job.
Colorfigures	Hand-inked isometric figures of everyday machines you can play with, by April Zhu. No AI involved	It's someone else's art, so we can borrow the format but not the drawings.	**Borrow the format** for a "Fig." series of UPSC machines (section on experience).

Pick	Verified facts	The case against	Verdict
Qwen-Audio-Jev paper	15 Oct paper: a decision head on an audio model that scores and classifies speech about 20 times faster than generating text	A paper, not a product we can call. Scoring a student's voice for mood would cross the app's own ethics rules.	**Later**, for interview-practice feedback such as pace and clarity.
Gist	On-device topic tagger: 36 fixed topics, 101 languages, 15 to 74 MB, free up to 100k devices	Its 36 topics are generic, not the UPSC syllabus. EmbeddingGemma, already planned, can map posts to the real syllabus.	**Skip.**

One trend worth naming: October brought a wave of **decision models** that return probabilities instead of text, including Cloudflare Clef (1 Oct), Clef-omni (9 Oct), Microsoft-Decision-1 and GPT-6 Luna Decisions (LLM Gateway timeline). That is the right tool for "which button to press", and it is why this design splits that job away from the chat model.

Where each AI is used

Each job gets the cheapest model that does it well, and only one of them writes Chotu's words.



Chotu's AI stack · five layers, one overnight job

A message goes from the phone to the Worker, which gathers the story card, the approved notes and the actions the decision layer allows, and only then calls the chat model. The overnight job is the only place the web is searched.

Surface	AI used	Runs where
Private notes: mood, journal	Gemma 4 on the phone; nothing is sent to the server	Phone
Chotu chat (Talk, Prep, Plan)	DeepSeek flash 1017, with the router falling back per language	Server to provider
Host mode in pods (digests, rituals)	DeepSeek flash in batch, off-peak	Server, scheduled
Next best step, routing, quiet-pod prompts	Clef-flash, plus rules	Workers AI
Notes search, pod matching	bge-m3 embeddings (already in the schema), EmbeddingGemma on the phone	Workers AI and phone
Daily notes and Chai cards	DeepSeek Search and Fetch, then DeepSeek flash drafts, then a person approves	Scheduled job
Reading PDFs and answer photos	Indra OCR for English PDFs and printed scans (English only for now)	A small Python box or Sarvam's API

Surface	AI used	Runs where
Voice	Bulbul V4 Flash live; Drama 3 for recorded English narration	Provider, cached in R2

Storage and memory

Everything stays on the Cloudflare pieces the server already uses (D1, R2, Durable Objects), plus Vectorize for search. The one new idea is the **Journey Ledger**: Chotu remembers events and a short story card, backed by the saved chat transcripts.

The tension to resolve: you want Chotu to know the student's backstory and journey, but the current spec keeps the context packet on the phone and says the server never stores it. My recommendation is to split by sensitivity rather than pick one side.

What	Where it lives	Who sees it	Kept for
Mood, journal, Close the Day	Phone only (as today)	Only the student	Until they delete it
Memory notes ("works till 6 pm")	Phone, with an opt-in encrypted copy in D1 for a second device	Student; editable in "What Chotu knows about me"	Until deleted
Journey events: attempt and age band, stage, target year, mocks taken, practice scores by topic, rhythm this week, pod joined, tools used	D1 `journey_events` table	Server and the decision layer	Account lifetime
Story card: about 300 tokens summarising the events in plain words	D1, rebuilt weekly and after big events (a mock, a result)	Shown to the student, who can edit or hide lines	Replaced each rebuild
Approved notes and Chai cards, each with source URL, date checked and an expiry date	D1 `notes`, embeddings in Vectorize, source PDFs in R2	Everyone	Until expired or superseded

What	Where it lives	Who sees it	Kept for
Nudge log: which action was offered, whether it was taken, what happened next	D1 `nudges`	Server only, used to learn which nudges help	180 days
Chat transcripts	D1 (older chats moved to R2), each message with an id	The student; the team only to check a complaint	Account lifetime; the student can export or delete
Pre-recorded audio and figures	R2, keyed by a hash of text, voice and language	Everyone	Until the text changes

Transcripts are kept as proof. Every story-card line, memory note and nudge reason stores the id of the message it came from. A "Why does Chotu think this?" tap shows that message, so we can always show that Chotu works only from what the student told it in the app, never from their social media or other browsing. The privacy policy should say exactly that.

How a turn is built, in this order so DeepSeek's cache does most of the work:

1. Chotu's persona and rules (the same for everyone, so it is almost always a cache hit at $0.003 per 1M tokens).
2. The action menu the decision layer allowed for this moment.
3. The student's story card and today's state (time in IST, Today's 3, days to Prelims).
4. The top 3 to 5 approved notes for the question, from Vectorize.
5. The message itself.

Mood and journal text never enter this packet, which keeps the current privacy promise intact.

Triggers that get each student to act

The app's real job is to catch the moment a student stalls and offer the smallest step that gets them moving. That needs a playbook of triggers. The AI's part is to spot the stall, pick the trigger that works for that person, and say it in their voice.

Stall moment	What the app sees	Trigger	Chotu says, for example
Opens the app and drifts	3 minutes open, nothing started	**Two-minute start**: one tap opens 5 questions on a weak topic	"Just 5 questions on Polity. Two minutes."
Usual study time passes	No activity by the time they chose at onboarding	**Their own if-then plan**, sent at their time: "After chai at 7, Today's 3"	"It's 7. Chai's done? Today's 3 are ready."
Plan feels too big	Today's 3 untouched by midday	**Shrink it** to a 20-minute version	"Let's do a 20-minute version of today."
Missed a day or two	No study events for 2 days	**Fresh start**, no guilt, pegged to a new week or month	"New week. Want a small restart plan?"
Studying alone, flat	Low activity, pod online	**Company**: the pod's study room or ritual starting soon	"4 from your pod are in the study room."
Reading without testing	3 or more notes read, no quiz	**Switch to recall** with a short quiz on what they just read	"You read 3 Polity notes. 5 questions to lock them in?"
After a bad mock	Mock score well below their average	**A small sure win**, then review only the 3 most useful mistakes	"Start with 5 you'll get. Then just 3 mistakes."
Lost in news and Telegram	Long news reading, no notes saved	**Today's Chai card**: the few items that matter, in 4 minutes	"Today's 3 news items that matter. 4 minutes."
Near a milestone	1 or 2 steps from finishing a unit	**Close the gap**: show how little is left	"2 quizzes and Polity basics are done."
Said they'd do something	A pod pledge is due	**Gentle accountability**: a reminder, then the pod's Friday check-in	"Your Chapter 5 pledge is due Friday. Want 30 minutes now?"
Still up late	Active after 23:30	**Stop**: Close the Day	"Past 11:30. Let's close the day."

How it runs:

1. Rules spot the stall from the Journey Ledger's events. That part is cheap and needs no AI.
2. Clef-flash scores the triggers that are allowed right now against the student's story card. Over time the nudge log teaches it what works for each person: some start for their pod, others for a tiny task.
3. Chotu writes the one line in the student's language and tone, and the tap opens the exact screen.
4. Success is measured as "started within 10 minutes", not time spent in the app.

Keeping triggers from wearing off

The table above is a starting set, not the answer. Like a parent's voice, any trigger fades with repetition, so the system has to keep measuring and keep trying new ones.

1. **Measure real lift, not clicks.** For each trigger and each student, the log compares how often they started within 10 minutes against a small holdout of moments where nothing was sent. A trigger only counts if it beats doing nothing.
2. **Track wear-off per person.** The lift is plotted by how many times that student has seen the trigger. When it falls to the no-nudge level, that trigger is rested for that person.
3. **Bring old ones back.** A rested trigger returns after a few weeks, because novelty often comes back with time. Its lift is then checked again.
4. **Keep exploring.** Most of the time Chotu uses what works for this person. About 1 time in 5 it tries something else, so the system never stops learning.
5. **Invent new triggers.** A weekly job asks the chat model for fresh variants based on what worked for similar students. It can change the messenger (Chotu, a pod-mate, a senior, a note from their own past self), the timing, the format (a question, a tiny task, a sound) or the wording. A person approves each one before it joins the playbook.
6. **Watch the long-term trend.** The weekly report shows which triggers work, on whom (by stage and attempt), and at what time of day. The goal it tracks is study days per week over a month, so triggers that win today but wear people out don't win.

None of this needs to be perfect at launch. What matters is that the nudge log records every trigger, the person and the outcome from day one, so the learning has data to start from.

The limits, from Chotu's spec and the community rules:

- How hard the app pushes is the student's choice (see Persistent nudges below). Gentle mode sends at most 2 a day, at times they picked. None between 23:30 and 07:00, and none for 72 hours after a result.
- No streaks, no rank comparisons, no shame, no fake urgency. The days-to-Prelims countdown appears in planning only, never as pressure.

- Every nudge has a "not today" button, and it is respected.
- If a student says they're in a bad place, nudges pause.

Persistent nudges, chosen by the student

A single ping is easy to ignore, so the app needs a nudge system that keeps at it until the student starts, the way Regain and habit games do. The student picks the strength at onboarding and can change it any time. Strict mode is opt-in, because a system that nags people who didn't ask gets switched off or uninstalled.

Mode	What happens when a study block is missed	Stops when
Gentle	One nudge at the chosen time	They start, or tap "not today"
Firm	A nudge, then a smaller ask 15 minutes later ("Just 5 questions"), then a pod-mate's or Chotu's check-in 30 minutes later	They start, or snooze with a reason
Strict	Everything in Firm, plus a pinned notification with the block's task and a timer that can't be swiped away. Opening Instagram or YouTube during the block shows a Chotu screen first: "You said 7 to 9 is Polity. Start 5 minutes, then decide."	The block ends, or they unlock with a reason, which is logged

How it works:

1. Each student's schedule and escalation run in a small per-student timer on the server (a Durable Object), which sends push notifications through Firebase.
2. The pinned notification and the app check run on the phone. On Android they need the student to grant usage access, and the Play Store asks for a declared reason. iOS needs Apple's Screen Time permission, which is a separate approval.
3. Each escalation step is itself a trigger in the learning loop above. If a step stops working for someone, it gets swapped like any other trigger.
4. Instead of streaks, which the spec rules out, the game-like hook is the week's rhythm ("4 of 5 days") and a shared pod goal that everyone's study minutes fill.

Admin: churn management

The admin console gets a churn page that shows who is slipping away early enough to act, and which nudges are pushing people out instead of in.

Panel	Shows	Lets the admin
At-risk students	Students whose study days per week are falling, who keep tapping "not today", who muted notifications, or who dropped from Strict to Gentle. Each gets a risk score from the decision layer, with the reasons listed	Send a pod-lead or senior check-in, or a fresh-start plan, with one tap
Gone quiet	Students with no activity for 7, 14 or 30 days, and uninstalls (detected when push delivery fails)	Run a win-back message, test two versions against each other
Retention by group	Day 1, 7 and 30 retention by signup week, stage, attempt and medium	Spot which groups leave and when
Trigger health	Each trigger's lift against the holdout, its wear-off, and the mute and uninstall rate after each nudge mode	Rest a trigger, approve new ones, cap a mode that drives people away
Why they left	Answers from a one-question exit prompt, plus Chotu's weekly summary of the patterns	Feed reasons back into the playbook

Not all churn is bad. A student who cleared the exam or chose a Plan B is a success, so the page tags them separately and never sends them win-back nudges.

How the experience levels up

The level-up is Chotu acting on the student's own journey, one well-chosen step at a time, rather than chatting better. Each idea below uses the layers above, and each can be measured in the nudge log.

1. **Next best step.** At the end of each session, the decision layer picks one action from a fixed menu of about 12: revise a weak topic, take a 10-question quiz, open Calm, join the pod's ritual, do Close the Day, sleep, see Attempt Compass. Chotu says it in one line with the reason ("You've missed Polity twice this week. 15 minutes?"). The menu, not the model, decides what is allowed, so Chotu can't invent a screen or push Plus.
2. **Chotu always has notes.** The daily job turns PIB, PRS, NCERT and newspaper items into approved notes overnight. Fact questions then get a sourced answer under the cautious rule you chose on 15 Oct, instead of "I don't have notes on that".

3. **A journey you can see.** Me shows the story card and a timeline: "Attempt 2. 217 days to Prelims. 41 quizzes, Polity up from 48% to 66%." Students correct it, which makes Chotu better and builds trust.
4. **Chai radio.** A 3-minute morning brief recorded once per language each day with Bulbul, stored in R2 and played by everyone. That costs pennies a day, not per student. Personal spoken replies are a "hear this" button, not the default.
5. **Photo of a Mains answer.** The student photographs a handwritten answer. Handwriting OCR reads it, and Chotu coaches structure (intro, body, way forward) against the directive word, without rewriting it for them. This is the feature aspirants pay coaching institutes for.
6. **The Fig. series.** Interactive isometric machines in the Colorfigures style, drawn as our own SVGs and checked by a person: how a bill becomes law, the monsoon engine, the RBI's repo lever, the GST pipe. These sit next to the existing Hidden Why explainers.
7. **Narrated classics in English.** Drama 3's directed delivery suits pre-recorded long reads, where a [warmly] or <slow> tag is worth paying for once.

The guardrails from Chotu's spec don't change: the decision layer never sees private notes, and nothing predicts ranks or tells anyone to quit.

Costs, risks and rollout

At 100k monthly users the whole AI stack costs roughly ₹70k to ₹95k a month, and voice and OCR are about half of it. These are my estimates at ₹85 to the dollar, using the usage assumptions in Chotu's spec (11 lakh chat turns a month).

Layer	Assumption	Per month at 100k users
Chotu chat (DeepSeek flash)	1,500 tokens in, 1,000 of them cached; 250 out; about ₹0.02 a turn off-peak, ₹0.04 at peak	₹22k to ₹44k
Decision layer (Clef-flash)	One call per session, about 800 tokens in	about ₹5k
Daily knowledge job (Search, Fetch, drafting)	About 500 calls a day	about ₹2k
Chai radio (Bulbul)	3 languages, 2,000 characters a day each	under ₹500
"Hear this" replies (Bulbul)	5% of turns, 300 characters each	about ₹33k
Answer photos (Sarvam OCR)	20,000 pages	about ₹10k

Most study hours fall off-peak: DeepSeek's peak is 06:30 to 09:30 and 11:30 to 15:30 IST on weekdays, so evening and weekend study gets the half price.

The risks I'd take seriously:

- **Free is not production.** The current NVIDIA and Gemini free routes have rate limits and terms that bar public apps. At ₹20k to ₹40k a month, paying for the chat lane is cheaper than the support load of a Chotu that times out.
- **Data hosting.** DeepSeek's own API is in China; Sarvam's copy of the same model is in India. That choice is yours, and the lawyer's DPDP check should cover it.
- **New-model churn.** 1017 is a day old. Keep the language-to-model router table so a lane can be swapped without code changes.
- **Wrong nudges.** Start the decision layer on rules plus Clef, log every nudge, and review the log weekly before letting it learn on its own.

Rollout, in order:

1. **Bake-off.** Run the 200-prompt eval in English, Hindi and Telugu on DeepSeek flash 1017 (direct and via Sarvam), Sarvam 105B and today's NVIDIA route. Pick per language on correctness, then speed, then cost.
2. **Knowledge job.** Search and Fetch plus Indra OCR fill approved notes daily, so Chotu always has sources.
3. **Journey Ledger.** The events table, the story card and the editable "What Chotu knows about me" screen.
4. **Decision layer.** The action menu, Clef-flash and the nudge log.

5. **Voice.** Chai radio first, then "hear this" with Bulbul; Drama 3 for narrated English content.
6. **Answer photos and the Fig. series.**
7. **Later.** vega-U for decisions on the phone and audio scoring for interview practice.